

safety in the chemical laboratory

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Waste Disposal in the Laboratory: Teaching Responsibility and Safety

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The U.S. Congress enacted the Resource Conservation and Recovery Act (RCRA) of 1976 to ensure the safe management of hazardous wastes. The goal was to address inadequacies in the management of industrial wastes and to encourage conservation and recycling measures for the reduction of some waste streams. After considerable delay, the U.S. Environmental Protection Agency promulgated regulations to implement RCRA, and these became effective November 19, 1980. The initial regulations identified particular hazardous waste streams and gave lists of hazardous chemicals and products whose presence required that a waste be treated as hazardous. In addition to the waste streams and substances listed, EPA indicated that wastes which are ignitable, corrosive, reactive, or toxic were also to be managed in accordance with these regulations. Waste generators were required to determine whether their wastes were hazardous (according to the EPA definitions).

While the regulations are long and complex, a quick look at the EPA definition of hazardous waste makes it clear that almost all

of the chemicals used in the teaching laboratories will ultimately become hazardous wastes. As laboratory instructors (and hazardous waste generators) we are responsible for the safe and proper disposal of laboratory wastes.

As is often the case with regulations, there is paperwork in the form of manifests and reports to assure that the generator can be identified and held responsible for the waste from "the cradle to the grave." Most of the record-keeping requirements have been eliminated for the small quantity generators who produce less than 1000 kg of hazardous waste per month. Though the quantities of waste generated in teaching laboratories is usually below this amount, the regulations still call for proper and safe disposal of the waste. In practical terms this means that you or an EPA-approved hazardous waste disposal firm must package the wastes in Department of Transportation (DOT)-approved containers for shipment to an approved disposal facility. Thus the disposal of hazardous waste becomes an added expense in the operation of a chemistry laboratory. This increasing cost should help motivate all those who teach chemistry to consider the wastes which are produced and to develop a program to minimize and manage chemical wastes. Ignoring the chemical waste program in the laboratory is not only an environmentally unsound practice, but places the laboratory instructor in jeopardy of prosecution under the EPA regulations.

The generation of potentially hazardous wastes occurs in almost every chemistry laboratory experiment. As a trained chemist you may realize that the wastes from a particular experiment are of low toxicity or hazard, but usually students do not have the background to make this judgment. Therefore, simply pouring wastes down the drain or putting them in regular trash cans without discussion will give your students the message that any chemical waste can be disposed of in this manner. By allowing laboratory wastes to be treated in a casual manner, we create the attitude that the environment will not be harmed because "the amounts are small," "it is diluted in the sewer or river," or "it is not very toxic." It is this type of attitude that has created pollution problems that resulted in

so much government regulation. The safe management and disposal of laboratory wastes sets a good example for students and trains them to be responsible scientists and citizens. Laboratory experiments should be designed with an eye to safe disposition of the wastes remaining. Students must be instructed about the hazards of the chemicals they use so they can cooperate in the collection of laboratory wastes. A safe disposal system including collection from laboratories, segregation of wastes and disposal or recovery is your legal and moral responsibility as chemistry instructors.

Generation and Collection of Laboratory Wastes

The prolific generation of wastes has become a part of our society, but the increased costs of materials and their disposal are beginning to change this trend. Any program of waste management for teaching laboratories must begin with an effort to maximize the use of chemicals and efficiency of experiments and to minimize the quantities and hazards of wastes produced. Laboratory exercises should be designed to engender in students an understanding of how safe and efficient research is conducted.

Most laboratory instructors already recognize the importance of safety considerations in the design and implementation of laboratory exercises. It should not be difficult to expand the concept of personal laboratory safety to the safe use and disposition of wastes generated in the laboratory exercise.

A first step is to minimize the quantities of reagents used. This can be accomplished by scaling down experiments. Besides reducing the amount of waste produced, scaled-down experiments can allow teaching new and valuable laboratory techniques using micro-scale equipment and can allow the introduction of new experiments for which the cost of chemicals would be prohibitive were macroscale equipment used. For example, in an organic laboratory, experiments can be designed around the use of 14/20 ground glassware rather than 24/40 glassware. In fact,

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some experiments at the University of Virginia use micro kits.

A second step in waste reduction is to minimize wastes resulting from excess use of reagents. While it is common to purchase chemicals in large containers to achieve cost savings, use of large containers in the laboratory can prove expensive in the long run. Students tend to take excesses of the reagent from the large containers only to have to throw it away later. If a large container becomes contaminated, large amounts of the reagent may be rendered useless. Large containers increase the likelihood of large spills which not only require time to clean up, but create an additional waste disposal problem. For these reasons, reagents should be transferred from large containers in a storage area to small containers for use in the laboratory.

In addition, as a part of teaching students the importance of preparation for laboratory exercises, it can be instructive to require students to know the quantity of the reagent needed for the experiment, safety precautions which must be taken in using the reagent, and the hazards of the wastes generated before the reagent can be issued to them.

Segregation of wastes should start at the point of collection. Students should be provided with clearly labeled containers and instructions for waste collection. The amount of segregation and number of containers will

depend upon the particular experiment. A minimum list of containers needed should include the following: (a) halogenated solvents (even if the experiment results in mixtures with non-halogenated solvents), (b) non-halogenated solvents, (c) aqueous solutions containing heavy metals of toxic compounds, (d) acids, (e) bases, (f) reactive compounds or solutions, (g) solid chemicals, (h) broken thermometers and mercury-contaminated waste. In cases where large quantities of a particular solvent are used, a large container should be provided for that solvent alone since segregation from other organic solvents makes distillation and recycling of the solvent possible. During any particular experiment only those waste containers that are actually required for that experiment should be placed in the laboratory.

The type of container and its location should be considered carefully. Special containers with clear signs designating the type of waste helps impress students with the seriousness of your concerns and efforts. The approach of using old reagent bottles for waste collection can lead to errors and fosters a careless attitude towards wastes. We have found it far more satisfactory to use 1-gal. or 5-gal., heavy plastic carboys with wide-mouth tops. The large-mouthed container allows liquids to be placed into the container without spillage (although it is still wise to put a tray or absorbent paper under the waste-collection containers). Clear or translucent containers allow students and instructors to see when the container is full. The use of relatively small containers is important to minimize the dangers of spills or fires involving large amounts of waste. Waste containers should be located conveniently for the

students but should not interfere with their experimental area. At the beginning of each laboratory period the location of the waste containers should be pointed out and the types of waste should be discussed. This discussion should include the consequences of mixing certain types of wastes which could cause dangerous reactions or fires. As the containers are filled or at the end of the laboratory period the waste should be moved to a suitable storage area away from the student laboratories and replaced by new or empty containers.

A secure area should be specified for storing the collected wastes. Because there are several possible dispositions for the wastes including recycling, neutralization, reuse, or shipment to an approved disposal facility, they should not be mixed. Since solvents can be recovered, they should be stored in a clearly labelled container and can be distilled for reuse in the laboratory or used as an auxiliary fuel. Wastes destined for commercial disposal must eventually be placed in Department of Transportation-approved containers for shipment. Since these containers are designed for safety, it is wise to accumulate and store wastes in the containers.

The storage area must be carefully managed. It should be kept uncluttered and wastes should be disposed of regularly to avoid the accumulation of large amounts of waste. Spill clean-up materials should be kept available to handle breakage or leakage in the waste storage area. (Note that any material used to clean up a spill is then deemed hazardous under EPA regulations.) Responsibility for the waste collection area should be clearly assigned. No waste should be allowed

in the collection area without proper labeling. Labels should include the composition of the waste (percentage of all known constituents), the experiment in which the waste was generated, and the date. The information is vital for proper disposal.

Disposal of Hazardous Wastes

The EPA requires generators of hazardous wastes to provide proper (safe and legal) disposal of laboratory wastes and to document the disposal. Although the EPA does not presently require small quantity (<1,000 kg/mo) generators to register as a hazardous waste generators, some waste disposal firms do require that their customers register. It is not the chemistry department alone, but the entire institution which will be registered as a hazardous waste generator. At institutions where there has been no organized program for hazardous waste collection and disposal, the total volume of waste may surprise the administration.

Most academic institutions will be forced to utilize an EPA-approved disposal company for at least some of their wastes. Commercial disposal costs are high and can be a significant factor in the economics of teaching chemistry. There are ways of keeping these costs at a minimum while maintaining good waste control practices and compliance with regulations. By using some chemical knowledge and common sense you may find reclamation, volume reduction, and neutralization of some wastes to be viable while using commercial disposal facilities for difficult or toxic wastes.

Since RCRA encourages resource recovery, recycling is the first waste reduction tech-

nique that should be considered. The wastes from each laboratory exercise should be evaluated to determine the value of recovery. The most obvious candidates are the organic solvents. This alternative to disposal will be most useful with the solvents used in the largest volumes. Segregation during the collection process helps to minimize the difficulties in redistilling mixtures. Redistillation does take time and effort, but it also reduces disposal costs and saves on the purchase of new solvents. Redistilling solvents and testing the purity of the product may even make an interesting laboratory exercise for a student or be assigned as a special project.

Another good candidate for reclamation is the silver from quantitative analysis laboratories. Several commercial products are designed to recover silver from photographic wastes, and the price of silver makes the effort worthwhile.

A number of different types of wastes can be returned to the manufacturer for reclamation. One example is waste mercury. Some commercial facilities will buy or take away used organic solvents to use as fuels. If this is to be done, wastes must be carefully segregated so they can be readily characterized. In most cases the quantities of solvents generated are too small to be of interest as fuels.

For those laboratory wastes deemed unworthy of recovery, volume reduction and neutralization should be considered. Since the objective is reduction or elimination of the hazards associated with the waste, any decisions about treating wastes should be made by trained chemists carefully considering the consequences of the treatment procedure. The most obvious candidates for

treatment are the acids and bases. While these wastes can be diluted to meet EPA standards (between pH 2–12.5), you may be required to prove that the solutions have been diluted before they go into the sink. When more concentrated acidic and basic wastes are collected separately, they can be carefully and slowly mixed together to neutralize their caustic natures. Remember to dilute and mix slowly to avoid heating and spattering. After neutralization and dilution, the solutions can be safely disposed of via the sanitary sewers. It is important to monitor the neutralization process to be sure that what goes down the drain has a pH in the range of 5–9. Chromic acid cleaning solutions should not be disposed of this way since the chromium in the waste is considered hazardous. You should also consider neutralization of reactive wastes, such as oxidizing and reducing agents. This does require some applied chemical knowledge, but the addition of an inexpensive reagent (or another waste) can make a solution safe for disposal through the sanitary sewers.

There are several useful sources of information about disposal methods. Some of the methods are applicable to the relatively small amounts of waste generated in teaching laboratories. Safety data sheets, available from the manufacturers of commercial chemicals, give some useful information. Another source of information is the *Aldrich chemical catalog* in which each listed chemical is coded to refer to the appropriate disposal table in the back of the catalog. The Manufacturing Chemists Association published the "Laboratory Waste Disposal Manual" in 1969, but this is not readily available as it has not been reprinted

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since 1972. The "NIOSH/OSHA Occupational Health Guidelines for Chemical Hazards" manual lists disposal procedures and many other safety guidelines. The "Toxic and Hazardous Industrial Chemicals Safety Manual" (International Technical Information Inst., Tokyo, 1978) gives handling and disposal information and a great deal of toxicity and hazard data.

In some experiments large volumes of relatively dilute wastes are generated. In the past many of these wastes probably went down the drain; strictly speaking, EPA does regulate these wastes. Pouring chemical wastes into a sink also sets a poor example for students who do not have the chemical knowledge to judge the safety of such practices. All laboratory generated solutions should be collected and a responsible chemist should decide the final disposition of the solution. Students should be instructed that sinks are to be used for disposal only when the instructor has specifically approved the practice for a particular solution or experiment.

Judgments about disposal via the sanitary sewerage system must be based upon consideration of the volumes, concentrations, and the regulations. It is difficult to imagine objections to pouring a few liters of dilute iron sulfate down the drain, but liters of arsenic waste could create bad publicity even if it did

not create a significant health hazard. Thus whenever wastes contain the elements arsenic, barium, cadmium, chromium, lead, mercury, selenium, or silver (for which EPA has given concentration limits) it is better to treat the solutions as hazardous even when concentrations are small. When volumes of such wastes are small, volume reduction should be considered. The type of volume reduction possible depends upon the nature of the chemicals involved. A large number of inorganic salts can be precipitated as hydroxides (using waste bases) or sulfides. The precipitates can be collected by filtration and the liquid neutralized and washed down the drain. Nitriles and cyanides can be converted to cyanates using sodium hydroxide and calcium hypochlorite.

When precipitation is not possible, concentration by evaporation should be considered. An inexpensive approach is to use plastic wash pans (to give a large surface area) to evaporate aqueous solutions in a fume hood. Where necessary the solutions should be neutralized before being evaporated. It is better to start with small volumes and add fresh solution whenever salts begin to form as this reduces the entrapment of liquid under a solid crust. The dried wastes can be packed in a labelled bottle and be included in a "lab pack" which is shipped for commercial disposal. One should avoid mixing more than one type of solution in the same evaporation container since this may lead to undesired reactions.

After minimizing the laboratory wastes by careful planning, recycling, neutralization, and volume reduction, a waste disposal problem still remains. If your entire institution qualifies as a small quantity generator,

you are relieved of record-keeping responsibilities but must dispose of your wastes in an approved manner. While the EPA regulations have helped insure the proper disposal of chemical wastes, they have also made it more difficult for generators with relatively small quantities of waste to contract for commercial disposal services. Many firms have a rather high minimum charge for the pick-up and disposal of wastes. If such is the case, you should consider contacting other institutions in your geographical area and arrange for several institutions to utilize the same disposal firm. By arranging for the disposal company to come to several institutions on the same trip you may realize significant savings.

It is not practical (or safe) to accumulate hazardous wastes over periods of greater than 90 days as a means of reducing the total disposal costs. The EPA regulations prohibit storage of hazardous wastes over 90 days unless you have a permit to operate a Treatment, Storage, and Disposal (TSD) facility. Thus one part of your waste collection procedure is to record the date that waste accumulation begins. There is some flexibility in the legal requirements because it is up to you to decide when a material becomes waste. Small quantity generators (who generate less than 1,000 kg/mo) lose their quantity status if they ever accumulate over 1,000 kg of waste. Thus laboratories must arrange for commercial disposal of wastes several times each year. The structure of the academic year can be to your advantage since some disposal firms will give a considerable reduction in their minimum charge if you arrange regularly scheduled pick ups. For example, one firm which charges \$1,000 as a minimum for chemical waste disposal reduces the minimum to \$500 if you schedule their services for 90-day intervals or for the end of each semester. At current prices the \$500 charge would cover the packing and disposal of four 55-gal. drums or "lab packs."

It is of utmost importance that you contact reputable disposal firms and investigate their services. Remember that any firm that takes your wastes must have an EPA (or state) permit to transport hazardous wastes and have a manifest system which allows you to prove that the final disposal of your wastes is safe and legal. The transporter must either operate or use a permitted TSD facility and you as generator must receive a copy of your manifest signed by the operator of the TSD facility within 35 days of shipment. If you do not receive this manifest, you should contact the EPA regional office near you and file an exception report to avoid any penalties.

Because of changing requirements (at disposal sites and in regulations) most disposal firms will only accept wastes that they pack themselves. To do this safely they need to know the contents of each container. Therefore your labelling system must be clear and informative. Writing over old labels is a dangerous and unacceptable practice. If you cannot tell them all of the chemicals in the waste with approximate concentrations, you may be required to have the waste analyzed (some firms provide this service at additional costs). Unknown mixtures make disposal more difficult and expensive. Record keeping is particularly important. Since both the generator and the disposal company have responsibilities for the waste, disposal firms are likely to require rather specific characterization of any wastes that they accept.

Certain types of wastes may not be accepted by the waste transporter and treatment facility at all.

When a disposal company packs your waste they may pour solvents into 55-gal. drums for incineration or recovery. Alternately they may make lab packs by filling open headed drums with individual bottles or containers and an absorbent material (like vermiculite). In each case segregation of materials is essential to avoid incompatible (and dangerous) mixtures of chemicals to minimize disposal costs, and to comply with DOT (Dept. of Transportation) regulations. Disposal firms know the regulations and know the disposal techniques that will be used so they should fill the shipping containers. For example, if they fill a 55-gal. drum with non-halogenated solvents that can be reclaimed or used as fuel, the disposal costs will be less than that for a similar drum which also contains chloroform or a known carcinogen. In the case of lab packs the final disposal will normally be burial so considerable absorbent must be used to avoid leaks from broken containers. There are strict DOT regulations about the size and type of containers (e.g., any containers packed inside the 55-gal. drum must be DOT approved if they contain more than 1 gal. of chemical). There are also compatibility requirements for packaging which should also be followed in collecting and storing your wastes. Do not store wastes (or chemicals) in alphabetical order but separate them into compatibility groups. Generally the bad (or forbidden) combinations include: (a) flammable liquids and oxidizers, (b) oxidizers and organic materials, (c) flammable liquids and corrosives, (d) poisons and corrosives, (e) acids and bases, and (f) water reactive chemicals and aqueous solutions. If you provide safe storage of your wastes, the disposal firm can worry about the DOT regulations for shipment. You should seek their advice on what type of containers to use for waste collection and accumulation as well as what information they need included on your waste labels. It is best to include input from your disposal firm as you design your program so costly mistakes can be avoided.

Laboratory waste disposal has entered a new age of complex regulations. While this clearly makes our jobs as chemistry instructors more difficult, there is a positive side. We can assure that waste chemicals will not be released to pollute our environment and we can seize upon this as an opportunity to teach our students more about safety, chemistry, and good citizenship.

